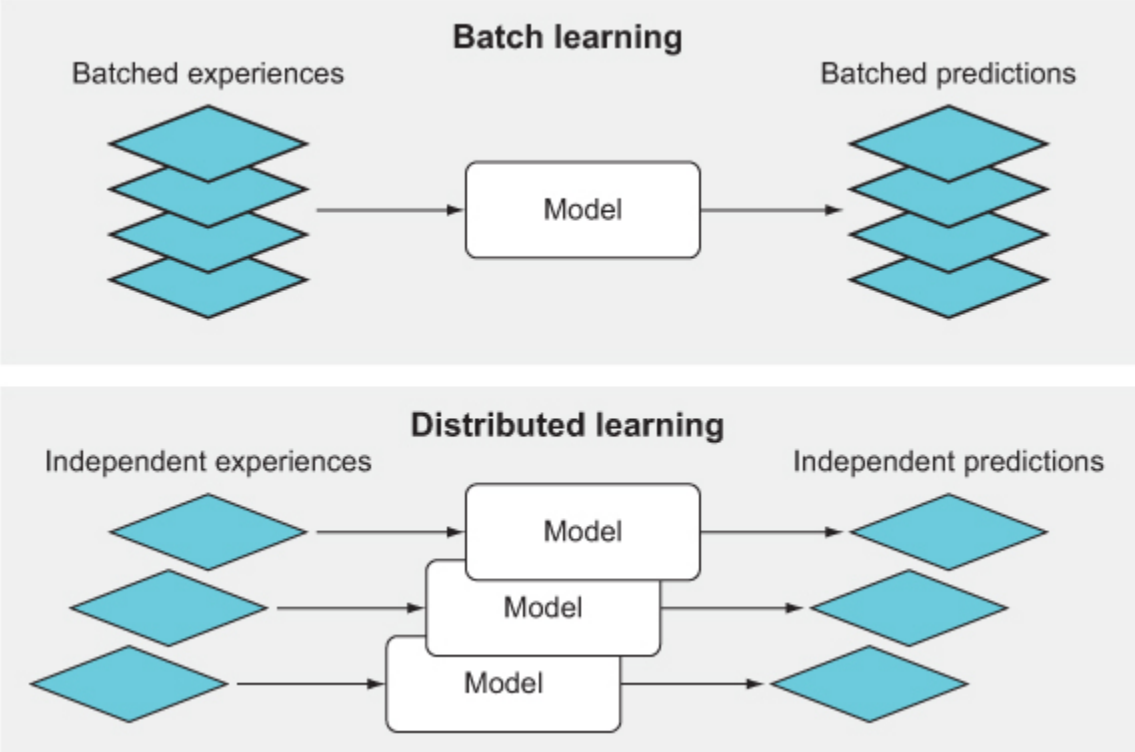


Multiprocessing versus multithreading

Modern desktop and laptop computers have central processing units (CPUs) with multiple cores, which are independent processing units capable of running computations simultaneously. Therefore, if you can split a computation into pieces that can be computed separately and combined later, you can get dramatic speed increases. The operating system software abstracts the physical CPU processors into virtual processes and threads. A process contains its own memory space, and threads run within a single process. There are two forms of parallel computations, *multithreading* and *multiprocessing*, and only in the latter form are computations performed truly simultaneously. In multiprocessing, computations are performed simultaneously on multiple, physically distinct processing units such as CPU or GPU cores.